Research Statement

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1. Research Background and Previous Work

I am completing a Master of Artificial Intelligence (Research Stream) at Monash University. My research focuses on watermarking and intellectual property protection for AI models, particularly watermark exposure and robustness under attack. I am interested in more than whether a watermark can be embedded and recovered. I also want to understand what an attacker can discover, how that knowledge might weaken the protection, and what evidence is needed to judge whether a method is reliable.

In my previous watermarking project, I carried out the core research work, including method design and implementation, experimental evaluation, data analysis and manuscript preparation. The resulting paper has been accepted at IEEE WIFS 2026. I am now working on a separate project that develops a reusable procedure for watermark detection and localisation in model parameters using statistical methods and frequency analysis. It also explores a watermarking method intended to balance stealth and robustness. I am preparing a manuscript on this work.

These projects have shaped how I approach security research. A method that works under a particular set of conditions may fail when an attacker changes strategy or gains more information. I have therefore come to value clear assumptions, fair comparisons and reproducible results. I enjoy turning an idea into an implementation, finding weaknesses through experiments, and revising my understanding in response to the evidence.

2. Future Research Interests

Model fingerprints and language model watermarks

I would like to explore model fingerprinting as well as watermarking for large language models. For model fingerprints, I am interested in whether the evidence used to identify a model or establish its origin remains reliable after updates, fine tuning or attacks. I also want to understand how a verification method can distinguish a genuine connection between models from coincidental similarities. Incorrect attribution matters, even when a method performs well on average.

For language model watermarks, I would like to study how reliable detection can be maintained without compromising text quality or usefulness. I am particularly interested in what protection remains after paraphrasing, editing or deliberate removal attempts. False positives and the interpretation of detection results are equally important to me. These questions build on my current interest in what watermark traces reveal and how to evaluate protection under attack.

Security and privacy of AI agents

My interests also extend to the security and privacy of models deployed as agents. When agents read external information, use tools, access sensitive data or collaborate with one another, their mistakes can affect actual operations rather than just the text they produce. I would like to understand how these systems can be manipulated or misused, and where useful safeguards can be introduced.

One question I would like to explore is how external content or messages from other agents influence an agent's decisions and subsequent actions. Could instructions hidden in a document or tool response cause it to depart from the user's goal? If one part of a collaborative system is manipulated, how does that influence spread through later interactions? I want to understand the conditions that make such attacks succeed, rather than only demonstrate that an attack is possible.

Privacy is another important part of this interest. Having access to information does not mean an agent should disclose it to every recipient or use it for every task. I would like to investigate how information sharing can be evaluated in relation to the user's intentions, the task and the parties involved. I am also interested in whether safeguards remain effective as these circumstances change during an interaction.

I would begin with clearly defined tasks that can be studied through repeatable experiments. My initial aim would be to understand attack mechanisms and the failure conditions of existing defences before exploring improvements. Evaluation should consider both the reduction in risk and the ability to complete legitimate tasks. A system that avoids harm by refusing most useful actions may offer little practical value. Likewise, success on a fixed test does not establish reliability across different interactions or attack strategies.

I see these interests as possible starting points for a focused doctoral project. Across them, I want to understand when evidence or protective mechanisms can be trusted and where their limits lie. Through further reading, experiments and discussion with my supervisor, I hope to develop a clear research question and a practical way to investigate it.

3. Research Approach and Long Term Goals

I want my research to explain why security problems arise and help reduce risks in practice. Dr Sahar Abdelnabi's work on agent security, privacy and safety evaluation interests me because it considers models in the context of their tasks and interactions. This is a perspective I would like to develop further. I hope to build on my experience of examining protection from both an attacker's and a defender's perspective while learning the methods needed to study these broader questions.

I am still at an early stage of my research career, but I take initiative and enjoy thinking creatively about problems. When I encounter a difficulty, I look for relevant information, try possible solutions and discuss what I find with collaborators. I am willing to take on additional work when it helps move a project forward. I enjoy proposing ideas, but I also want to test them carefully and change them when the evidence calls for it. In a team, I aim to complete my responsibilities on time, share ideas and help identify problems that might otherwise be missed.

My longer term goals are not tied to a single technique. I believe advances in AI capability should be accompanied by serious attention to security and privacy, rather than leaving these concerns until systems are already widely used. Through doctoral study, I hope to develop the independence and judgement to ask useful questions, investigate them rigorously and produce work with practical significance.